

SKILL WEEK 6

Task 1:

Code:

```
#include <stdio.h>

int main() {
    char input[200];
    int i;

    printf("Enter a string with escape characters: ");
    fgets(input, sizeof(input), stdin);

    printf("\nProcessed string:\n");
    for (i = 0; input[i] != '\0'; i++)
    {
        if (input[i] == '\\' && input[i + 1] !=
            '\0') {
            i++;
            switch (input[i])
            {
                case
                'n':
                    printf("\n");
                    break;
                case 't':
                    printf("\t");
                    break;
                case ' ':
                    printf(" ");
                    break;
                case '\\':
                    printf("\\");
                    break;
                case '$':
                    printf("$");
                    break;
                case '!':
                    printf("!");
                    break;
                default:
                    printf("%c", input[i]);
            }
        }
    }
}
```

```

    } else {          printf("%c",
input[i]);
    }
}

return 0;
}

```

```

charanteja_a@charan:~$ nano exp6.1.c
charanteja_a@charan:~$ gcc exp6.1.c -o exp6.1
charanteja_a@charan:~$ ./exp6.1
Enter a string with escape characters: hi

Processed string:
hi
charanteja_a@charan:~$ |

```

Task 2:

Code:

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {

    pid_t pid;

    pid = fork();
    if (pid < 0) {
        perror("fork failed");    return
1;
    }

    if (pid == 0) {

        // Child process
        printf("Child
process created.\n");
        printf("Child
PID: %d\n", getpid());

        char *args[] = {"ls", "-l", NULL};    execvp(args[0], args);
    }
}

```

```
        // This runs only if execvp fails
        perror("execvp failed");    exit(1);

    } else {

        // Parent process    printf("Parent
        process.\n");    printf("Parent PID:
        %d\n", getpid());

        wait(NULL);

        printf("Child process finished.\n");
    }

    return 0;
}
```

Output:

```
charanteja_a@charan:~$ nano exp6.2.c
charanteja_a@charan:~$ gcc exp6.2.c -o exp6.2
charanteja_a@charan:~$ ./exp6.2
Parent process.
Parent PID: 1085
Child process created.
Child PID: 1086
total 872
drwxr-xr-x 12 charanteja_a charanteja_a 4096 Aug 25 03:38 OSSP
-rwxr-xr-x  1 charanteja_a charanteja_a 16384 Aug 31 05:09 client
```